

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 8: Appendix

by

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix describes the evaluation pipeline used to generate the numbers in Chapter 4, including the data sources, the processing steps, the default parameters, the formula for computing proxies, and a glossary. These quantities are also repeated in the tables and figures in Chapter 4.

8.1.1 Pipeline stages

The evaluation consists of two stages, performed on Arbitrum One during the observation period 2025-12-01 to 2026-05-31 (UTC, both inclusive).

1. Reputation graphs and scores: Fetch ERC-20 Approval and Transfer logs in monthly chunks (per-query dry-run, and a 150 GiB abort limit). The set of wallets included in our sample is those with non-zero approval or transfer connectivity ($n = 5,521$). Construct the latest-allowance endorsement graph and the time-decayed transfer graph, then calculate EndorseRank and AWP scores using power iteration (damping $d = 0.85$, tolerance 10^{-8} , up to 300 iterations); compute C-PR on the per-node mixture of the two layers with $\lambda \in \{0.25, 0.5, 0.75\}$ and S-PR on the transfer layer with the EndorseRank restart vector, also using power iteration (same solver parameters).
2. Checks, alignment, and benchmarks: Calculate transfer and allowance checks from Chapter 3, report Spearman ρ and Kendall τ against each score vector with 95% paired-bootstrap confidence intervals (400 resamples of wallets, seed 42), the five predefined single-layer $\Delta\tau$ differences and the eight hybrid $\Delta\tau$ differences, measure PageRank solve times on pre-built edge lists under one protocol (one unmeasured warm-up run, five measured runs, per-session memoization), and perform sensitivity analysis over damping factor, number of top tokens,

and sample definition.

3. Temporal holdout: Freeze scores at 28 February 2026 on the spender cohort ($n = 1,335$), count new approval pairs and new transfer senders in March–May 2026, and report every score and both raw-degree baselines against both labels with 400 paired resamples and the eight pre-registered holdout contrasts (`scripts/run_holdout_eval.py--cohortspenders`).

The tables of Chapter 4 were generated by `scripts/run_dissertation_eval.py--real--robustness--export-latex--benchmark-repeats5` and `scripts/run_holdout_eval.py--cohortspenders` on 14 September 2026, then written to `results/tables/` from the machine-readable summaries. Those summaries (`eval_summary.json`, `holdout/eval_summary_senders.json`) and a copy of the configuration in force are committed under `A-Skill-Programs/margin_rank/results/`, so every number in the tables can be checked against the summary without re-running the pipeline. The evaluation source (`A-Skill-Programs/margin_rank`), the versioned configuration (`config/margin_config.yaml`) and the summaries are in the public repository https://github.com/abitocodes/phd_works at revision [revision-pending-commit]. The coupling weights, holdout labels and success rules were written to the configuration and to the plan of 14 September 2026 before the coupled scores were computed.

Offline synthetic checks use the same scoring and alignment logic on fixture graphs (no cloud access) to validate implementation before real extraction. Real extraction requires Google Cloud credentials and a billing project; scans target only the public Arbitrum One log dataset.

8.1.2 BigQuery extraction design

ERC-20 Approval and Transfer events are identified by their standard topic-0 hashes and restricted to logs whose owner/spender or from/to address lies in the evaluation wallet list, so scan cost scales with the cohort rather than the full chain. Monthly batching keeps each query under the operational byte cap. Representative SQL patterns appear in Appendix 8.2.

Extraction records (bytes processed, row counts, timestamps) are retained in the study’s acquisition

log for auditability; this research reports the resulting cohort sizes and edge counts rather than those operational logs.

A second-tier extraction was planned for the 93,559 supplemental addresses of the expanded wallet pool ($N = 99,080$; Section 3.12.1). The dry-run estimate for the six monthly approval and transfer queries over that address set exceeded 1.3 TiB, and the download stage failed repeatedly on the storage-backed result path; the extraction was therefore abandoned within the study period. The supplemental addresses consequently enter the expanded-pool experiments as nodes only, with no approval or transfer edges of their own, which is why the full-pool row of Table 4.2 reproduces the matched-cohort graph. Completing that extraction is listed as future work in Chapter 6.

8.1.3 Scaling tables

Table 8.1 reports in-cohort runtime scaling for $n \leq 5,521$ under the timing protocol of Section 3.11; its last row is the same measurement as Table 4.1. All of the in-cohort stages are subsets of the wallets whose edges were extracted. Unlike the expanded-pool stages, $|V|$ and $|E|$ therefore grow with n here.

Table 8.1: In-cohort runtime scaling (SHA256 seed benchmark-scale-v1; $n \leq 5,521$ matched GMX cohort). Here both $|V|$ and $|E|$ grow with n because every stage is a subset of wallets whose edges were extracted. Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs.

n	ER (s)	AWP (s)	Speedup	ER $ V $	ER $ E $	AWP $ V $	AWP $ E $
1,000	0.056	0.502	9.0×	1,302	2,584	6,463	23,058
2,000	0.170	2.055	12.1×	2,362	5,147	11,849	47,031
4,000	0.368	3.366	9.2×	4,645	10,557	22,591	97,105
5,521	0.401	4.105	10.2×	6,442	14,727	30,791	137,087

8.1.4 Credit-delegation extract (deferred)

Aave V3 BorrowAllowanceDelegated events were not extracted in this study. Any ranking that depends on credit-delegation edges is therefore undefined on the present sample and is reported as

non-interpretable ($\tau = \text{---}$) in the archived Aave-oriented matrix (Appendix 8.4). Extending the study to delegation-based reputation would require a dedicated extract of those events and a fresh alignment evaluation; it is left to future work.

8.1.5 Supplementary alignment summary table

Table 8.2 lists family-mean τ for all six proxy families computed by the pipeline, including the liquidation, inverse-risk and GMX trading-success families that were withdrawn from the primary comparison together with the GF-PR reference (Section 2.8). The transfer, allowance and Sybil-stability rows are those of Table 4.3.

Table 8.2: Mean Kendall τ by proxy family: EndorseRank and AWP ($n = 5,521$).

Family	EndorseRank	AWP	Higher
Transfer	0.333	0.534	AWP
Allowance	0.355	-0.030	EndorseRank
Liq.	0.093	0.113	AWP
Inv.risk	0.035	0.107	AWP
Sybil	0.233	0.286	AWP
GMX	0.096	0.179	AWP

8.1.6 Parameter summary

Table 8.3: Default parameters used in the primary evaluation.

Parameter	Value	Role
PageRank damping d	0.85	Default teleport mixture
Convergence tolerance ε	10^{-8}	Power-iteration stop
Max iterations	300	Power-iteration cap
Logistic k	0.01	AWP decay slope
Logistic t_0	180 days	AWP decay midpoint
Coupling weight λ	0.5 (primary); 0.25, 0.75	C-PR allowance-layer weight, fixed before the run
S-PR restart vector	normalised $\mathbf{s}_{\text{ER}}$, floor 0	Endorsement-seeded teleportation
Holdout freeze / outcome window	2026-02-28 / 2026-03-01 – 2026-05-31	Spender cohort $n = 1,335$
Observation window	2025-12-01 – 2026-05-31	UTC inclusive
Bootstrap resamples / seed	400 / 42	Paired wallet bootstrap, percentile 95% CI
Timed repeats / warm-up	5 / 1	Runtime protocol
In-cohort subsample seed	benchmark-scale-v1	Deterministic SHA256 seed
Expanded pool subsample seed	benchmark-tier2-v1	Deterministic SHA256 seed

8.1.7 Proxy operational definitions (reference)

The following formulas match Chapter 3.

- Transfer: $\text{in_degree}(w) = |\{u : \exists u \rightarrow w\}|$; $\text{in_value}(w) = \sum_{e:u \rightarrow w} \text{value}(e)$.
- Allowance: $\text{in_approve_degree}(w) = |\{o : \exists o \rightarrow w\}|$; $\text{in_approve_value}(w) = \sum_o w_a(o, w)$ with w as spender.
- Sybil-adjusted stability: $\text{inbound_counterparty_ratio}(w) = |\{u : u \rightarrow w\}|/|T^{\text{in}}(w)|$ (self-transfers excluded); tenure in days; active months.

8.1.8 Glossary

Allowance ERC-20 authorization for a spender to transfer tokens on behalf of an owner.

Alignment Rank agreement between reputation scores and a validation proxy (ρ, τ) .

Construct overlap Shared data source between a ranking method and a proxy, inflating alignment.

Endorsement graph Directed graph of latest allowances (EndorseRank input).

Matched cohort Primary sample of $n = 5,521$ wallets with non-zero endorsement- and transfer-graph connectivity.

Expanded wallet pool Larger address set ($N = 99,080$) used for node-count sensitivity of runtime; supplemental addresses have no extracted edges.

Paired bootstrap Resampling wallets with replacement using one shared index matrix for every coefficient, so that differences between coefficients can be tested on the same resamples.

PageRank Spectral ranking by stationary distribution of a damped random walk.

Proxy Observable wallet metric used for domain-intuition validation.

Teleport Uniform jump in PageRank with probability $1 - d$.

8.2 SQL Templates, Configuration, and Extended Notes

This appendix collects representative SQL patterns, configuration keys, and extended methodological notes that support reproduction.

8.2.1 Approval log extraction pattern

ERC-20 Approval events are identified by topic 0 equal to the keccak256 hash of Approval(address,address,uint256). A simplified monthly query pattern is:

```
SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS owner_topic,
    topics[SAFE_OFFSET(2)] AS spender_topic,
    data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
    AND block_timestamp <  TIMESTAMP('2026-01-01')
    AND topics[SAFE_OFFSET(0)] =
    ↪ '0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925'
    AND (
        LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
        ↪ UNNEST(@wallet_list)
        OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
        ↪ UNNEST(@wallet_list)
    );
```

Wallet filtering via @wallet_list keeps scans proportional to the cohort rather than the full chain. Monthly batching respects per-query byte caps documented in the study’s acquisition record.

8.2.2 Transfer log extraction pattern

Transfer events use topic 0 for Transfer(address,address,uint256):

```
SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS from_topic,
    topics[SAFE_OFFSET(2)] AS to_topic,
    data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
    AND block_timestamp <  TIMESTAMP('2026-01-01')
    AND topics[SAFE_OFFSET(0)] =
    ↪ '0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'
    AND (
        LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
        ↪ UNNEST(@wallet_list)
        OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
        ↪ UNNEST(@wallet_list)
    );
```

Post-processing applies token decimals, builds time-decayed edge weights, and restricts nodes to the evaluation sample before PageRank.

8.2.3 Configuration keys

The pipeline’s parameters are collected in a single versioned configuration file. Salient keys include:

- observation window start and end timestamps;
- PageRank damping, tolerance, and maximum iterations;
- logistic decay k and t_0 ;
- benchmark repetition count and subsample seeds;
- output locations for intermediate artifacts and table fragments (results/tables/).

If the damping factor or the decay parameter are changed, then the entire evaluation has to be run again (including robustness sweeps and exporting the tables) and all figures re-created from the same artifacts so that the numerics match what is reported in the pipeline’s machine-readable summary.

8.2.4 Complexity notes

Let n be the number of nodes and $m = |E|$ the number of edges. Constructing the latest-allowance edges takes $O(R_a \log R_a)$ time if the R_a approval rows are already sorted by timestamp, or $O(R_a)$ expected time using a hash map keyed on (token, owner, spender). Constructing the transfer edges takes $O(R_t)$ time over the transfer rows. PageRank runs in $O(m + mI)$ time for I iterations, where the first term is the cost of building the sparse operator. The iteration count and edge counts for both methods are shown in Table 4.1. EndorseRank uses about one-third the number of iterations that AWP does and has $m_{\text{approve}} \ll m_{\text{transfer}}$, which means that both aspects work in EndorseRank’s favor.

Memory usage is dominated by the edge lists and score vectors. The peak-memory ratio in Table 4.1 is on the same order of magnitude as the edge-count ratio.

8.2.5 Conventions for the result tables

The tables of Chapter 4 follow these conventions:

1. Indicate whether a row corresponds to a social method (EndorseRank or AWP).
2. Associate each family with its target concept (allowance vs. transfer).
3. Favor relative comparisons on the same cohort rather than absolute τ thresholds.
4. Treat runtime tables as representing PageRank solve time on pre-built edge lists, not end-to-end ETL time, and assume that the numbers will be the same whenever the same graph is reported in the same configuration (see Section 3.11).
5. Report every τ with its interval and derive differences between coefficients from Table 4.4 rather than subtracting the point estimates.
6. Treat the sample-definition sweep (Table 4.10) as a sensitivity analysis and do not compare its absolute values to the matched-cohort tables.

8.2.6 Archived extended baselines

The three-method (including the now-withdrawn GF-PR reference), seven-method and six-Aave alignment matrices have been archived with the evaluation pipeline, along with the corresponding CSV matrices. They are excluded from the main comparison because some variants rely on additional event types (such as credit delegation) or introduce concepts that are not part of the EndorseRank–AWP question.

8.2.7 Limitations

The SQL templates do not mean that other chains were queried, nor do the archived baselines serve as peer comparisons in this study. The numerics presented are those in the pipeline’s machine-readable summary for the matched cohort ($n = 5,521$) and the expanded wallet pool ($N = 99,080$).

8.3 Elementary Properties of PageRank and Rank Correlation

This appendix summarizes the elementary properties used implicitly throughout Chapters 2–4. This material is standard ^{[15],[20],[21],[23]}.

8.3.1 Existence and uniqueness of PageRank

Let P be a row-stochastic matrix on n nodes (possibly with dangling-node repair as in Equation (2.1)). For $d \in (0, 1)$ the Google matrix

$$G_{\text{PR}} = dP^\top + \frac{1-d}{n}\mathbf{1}\mathbf{1}^\top$$

is positive and column-stochastic (up to the transpose convention of Equation (2.2)). From the Perron–Frobenius theorem for positive matrices, G_{PR} has a single eigenvalue of modulus one and a corresponding eigenvector π that is strictly positive, and we can normalize it such that $\sum_i \pi_i = 1$. So there are unique PageRank scores for any directed graph, assuming fixed damping and dangling-node rules ^[23].

Uniqueness ensures a well-defined comparison between EndorseRank and AWP: with fixed parameters, each method defines only one score vector on a given edge set. Hence differences between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ must stem from edge semantics and weights rather than multiple solvers.

8.3.2 Continuity in edge weights

PageRank is a continuous function of the entries of P for fixed $d \in (0, 1)$. Allowing for minor variations in the values of the allowance and transfer weights will have only a small effect on scores in the $\|\cdot\|_1$ norm, with Lipschitz constants dependent on d and n ^[23].

Continuity enables us to carry out robustness checks by changing damping or limiting to the top tokens. If alignment was overly sensitive to the exact hyperparameters, such small changes would disrupt the family-level τ altogether. Empirically, EndorseRank’s allowance family mean is unchanged to three decimal places across $d \in \{0.75, 0.85, 0.95\}$ (Table 4.8), suggesting a stable

ranking geometry.

8.3.3 Sparsity and per-iteration cost

A sparse matrix–vector product $P^\top \mathbf{x}$ takes $O(m)$ arithmetic operations for m nonzero edges, and constructing the sparse operator from the edge list also takes $O(m)$ time. So power iteration takes $O(m + mI)$ time for I iterations. The transfer graph on the matched cohort has around 9.3 times as many edges as the endorsement graph, meaning the transfer method would take approximately an order of magnitude longer even with identical I . The gap is further widened by different iteration counts and the measured wall-time ratio (Table 4.1) is on the same scale as the edge ratio.

8.3.4 Properties of Spearman’s ρ

Spearman’s ρ is Pearson’s correlation applied to the ranks. It is invariant to strictly increasing transformations of either variable and takes values in $[-1, 1]$. $\rho = 1$ indicates perfect monotone agreement and $\rho = -1$ indicates perfect reverse agreement. Ties are handled via mid-rank adjustments; the evaluation pipeline uses standard tie-aware implementations. Due to the continuous and heavy-tailed nature of reputation scores and proxies, rank-based association is more appropriate than Pearson correlation on raw values. The AWP baseline paper similarly evaluates its ranking using rank statistics [3].

8.3.5 Properties of Kendall’s τ

Kendall’s τ is the normalized difference between the number of concordant and discordant pairwise orderings. It can be understood probabilistically: for two randomly chosen distinct wallets, τ is the probability their relative ordering is the same under both rankings, minus the probability their relative ordering is different [21]. A τ in the mid-0.3 range, like EndorseRank on the allowance family (Table 4.3), implies a substantial but imperfect agreement between the two rankings.

For large n , τ and ρ tend to have similar signs and magnitudes, though they are not the same. We report both, following the AWP baseline [3], and we use the family-mean τ with a paired-bootstrap

interval ^[34] as our main summary statistic.

8.3.6 Construct validity note

Cronbach and Meehl ^[22] distinguish between criterion-related validity and construct validity. Without a definitive gold standard default label, this research uses a construct-oriented approach: each method is tested against validation checks that embody its conceptual definition (e.g. receiving allowances for EndorseRank; making inbound transfers for AWP), as well as a holdout period of future new approvals. High within-construct τ supports interpretation. Low cross-construct τ is not automatically failure; it may indicate discriminant validity.

8.3.7 Where each primary quantity is reported

The following index gives the table in which each primary quantity is established.

Quantity	Reported in
Matched-cohort size, $ V $, $ E $, runtime, SD, memory, iterations	Table 4.1
In-cohort scaling stages	Table 8.1
Expanded-pool stages ($N = 99,080$)	Table 4.2
Family-mean τ with 95% CI; inter-method τ	Table 4.3
Paired $\Delta\tau$ contrasts	Table 4.4
Per-proxy ρ , τ , CI	Tables 4.5, 4.6, 4.7
Damping, top-token, sample-definition sweeps	Tables 4.8, 4.9, 4.10
Holdout spender cohort ($n = 1,335$)	Table 4.13
Observation window	2025-12-01 – 2026-05-31 (Table 8.3)

The quantities listed above appear in the evaluation summary and in the tables of Chapter 4.

8.4 Archived Extended Baselines

Additional PageRank variants were scored on the same matched cohort and proxy families. They are not part of the primary EndorseRank–AWP research questions.

8.4.1 Seven-method alignment matrix

Seven-method matrices remain in the evaluation archive (LP-PR, CW-AWP, LF-PR, and RiskProp-style variants). They are not drawn here. EndorseRank remains the allowance-family leader among social methods; AWP (and closely related transfer variants) lead on transfer.

The archive matrices show that:

- Methods built on transfer-like edges cluster near AWP on transfer and Sybil families.
- Methods with very small edge counts can produce unstable or degenerate rankings; their run-times may be low simply because m is tiny.
- RiskProp-style rows that mirror AWP numerically indicate shared transfer structure, not independent discovery.

8.4.2 Six-method Aave-oriented matrix

An Aave-oriented preset remains in the evaluation archive. Several lending-event graphs are sparse on Arbitrum within the study window; delegation-based rows may be degenerate when BorrowAllowanceDelegated events were not extracted (Appendix 8.1).

The Aave-oriented preset remains in the evaluation archive and is not drawn here.

These matrices motivate future work on credit-delegation edges and cross-protocol lending labels, but they do not alter Claims C1–C4.

8.4.3 Why the comparison uses two methods

The primary comparison is EndorseRank versus AWP for three reasons:

1. The novel construct is allowance-based endorsement. Comparing it to the established transfer baseline ^[3] isolates the research question.
2. Additional variants that share transfer edges with AWP add rows without adding independent constructs.
3. The outcome-native GF-PR reference used in the research proposal is built from the same

GMX events as the trading-success proxies, so any alignment between them is construct overlap, not validation (Section 2.8); its three-method matrix is archived alongside the seven-method and six-Aave presets.

8.4.4 Reproducing archive tables

Archive tables are regenerated by the evaluation pipeline’s table export with seven-method and six-Aave presets. CSV copies are recorded alongside the table fragments.

8.4.5 Notes on archived method identifiers

The following identifiers appear in archive tables and CSV exports.

LP-PR A transfer-graph variant with alternative loss-penalized or liquidity-oriented weighting. Empirically it tracks AWP closely on transfer families, which points to shared payment-flow structure.

CW-AWP A counterparty-weighted AWP variant that down-weights repeated transfers between the same pair. Slightly lowers transfer τ vs. AWP; same construct family.

LF-PR A low-frequency or lightly filtered graph; very small edge counts, near-zero runtimes. Rankings not comparable to full AWP: the sparse graph is a failed baseline.

RiskProp A risk-propagation style ranking on transfer-like edges, inspired by Lin et al. ^[12]. On this cohort it numerically mirrors AWP on several families, which implies the risk signal implemented did not diverge strongly from transfer centrality under the study’s parameterization.

LiqCall-PR / Borrow-PR / Repay-PR Aave-oriented wallet-to-wallet graphs based on liquidation calls, borrows, and repays (where applicable). Edge counts on Arbitrum in the study window are small (tens to hundreds), so alignment is weak and runtimes are negligible.

Delegation-PR Intended to leverage credit-delegation allowances; with zero delegation edges extracted, scores are degenerate and τ is not interpretable.

Archive rows are not peer-reviewed baselines.

8.4.6 Criteria for the two-method comparison

Chapters 4–5 focus on EndorseRank and AWP. The other variants do not meet the following criteria:

1. Construct independence: a baseline should implement a different edge semantics, rather than simply being a reweighting of transfers.
2. Data adequacy: the edge counts should be large enough to support stable PageRank on the matched cohort.
3. Scope: the contribution is allowance-based endorsement, not a survey of all possible PageRank variants.

Based on these criteria, EndorseRank (allowance) and AWP (transfer) form a minimal sufficient set. The seven-method and six-Aave matrices remain in the archive; they are not included in Claims C1–C4.

8.5 Claim-to-Evidence Map

This appendix links each hypothesis, research question, and empirical claim to the corresponding evidence in Chapter 4.

8.5.1 H1 / RQ1: Rank divergence

- Statement: EndorseRank and AWP produce different wallet rankings on the same cohort.
- Evidence: inter-method Kendall τ with 95% CI (Table 4.3, last row; Section 4.5); rank scatter plot in Figure 4.4; score distribution plots in Figure 4.3.
- Interpretation: related but non-equivalent rankings; O1 criterion is satisfied (Table 5.1).

8.5.2 H2 / RQ2: Construct-specific alignment

- Statement: the two methods yield different results on allowance vs. transfer checks at the same time points.
- Evidence: family means with CI (Table 4.3); paired $\Delta\tau$ contrasts ii and iii (Table 4.4); per-proxy detail (Tables 4.5 and 4.6); heatmap in Figure 4.2.
- Note: contrasts i and v in Table 4.4 contain zero; the claim applies only to between-method comparisons.

8.5.3 H3 / RQ3: Computational efficiency

- Statement: EndorseRank is faster than AWP at the same n , due to fewer edges.
- Evidence: Table 4.1 (runtime, SD, memory, iterations, $|V|$, $|E|$); Table 8.1; Table 4.2; Figure 4.1.
- Note: expansion steps add nodes without adding edges (Section 4.2.1); the claim is restricted to edge count.

8.5.4 RQ4: Temporal holdout

- Statement: after scores are fixed at t_1 , EndorseRank predicts t_2 new owner–spender approvals on inbound spenders better than AWP; it does not predict them better than the t_1 in-approve degree.
- Evidence: Tables 4.13 and 4.14 (Section 4.8): EndorseRank minus AWP is greater than zero; EndorseRank minus t_1 in-approve degree is less than zero; AWP minus t_1 in-degree on new transfer senders is less than zero.
- Qualification: both labels are the constructs in the future tense (new approvals, new senders), not credit or trading outcomes, and both are counts of arrivals; RQ4 replaced the proposal’s trading-success comparison (Section 2.8).

8.5.5 RQ5: Coupled operator

- Statement: a per-node convex mixture of the allowance and transfer layers (C-PR) does not rank wallets better than the appropriate single-layer method; the seeded ablation (S-PR) reproduces AWP.
- Evidence: Table 4.14 (pre-registered primary rule, both parts not met); Tables 4.11 and 4.12 (same-window position, secondary rule); runtime rows in Table 4.1.
- Qualification: fixed $\lambda \in \{0.25, 0.5, 0.75\}$, uniform teleportation, transition-level mixing, one chain; the rule and the values of λ were fixed in configuration before the run (Section 3.7).

8.5.6 Claims C1–C5

- C1: runtime evidence as in H3/RQ3; specific to EndorseRank, since C-PR costs at least as much as AWP (Table 4.1).
- C2: intended-construct checks (allowance for EndorseRank, transfer for AWP) in Table 4.3.
- C3: construct-specific winners as in H2/RQ2, tested in Table 4.4.
- C4: efficiency–alignment trade-off combining C1, C3 and the spender holdout (Tables 4.13 and 4.14), bounded by the degree baselines; robustness in Tables 4.8–4.10 and Figure 4.5.
- C5: bounded negative result on coupling, as in RQ5.

8.5.7 Explicit non-claims

This research does *not* claim:

- that AWP is obsolete;
- that EndorseRank’s allowance alignment exceeds its own transfer alignment;
- that EndorseRank or AWP predicts the future growth of its own construct better than the raw t1 degree;
- that no combination of the two edge types can beat the single-layer methods; only that the tested per-node convex mixture with fixed λ does not;

- that runtime scales favorably as the edge population grows beyond the cohort graph;
- that results generalize beyond Arbitrum One and the study window without replication;
- that on-chain scores replace collateral or legal identity systems.

8.5.8 Replication artifacts

1. Merged wallet-ranking table: scores for EndorseRank, AWP, C-PR ($\lambda \in \{0.25, 0.5, 0.75\}$) and S-PR.
2. Machine-readable holdout summary for the spender cohort, including the paired contrasts and the primary-criterion verdict (committed under `results/holdout/`).
3. Machine-readable evaluation summary: authoritative same-window numerics, per-run timings, bootstrap protocol (committed under `results/`).
4. Exported table fragments (`results/tables/`).
5. Exported vector figures (`Figures/generated/`).
6. Optional archived extended-baseline matrices.

Tables and figures are regenerated by running the evaluation pipeline on real data with robustness sweeps, exporting the tables, and regenerating figures from the same artifacts (Appendix 8.1.1).

8.6 Ethics Clearance Certificate

8.6.1 Certificate

This study uses public blockchain data only and involves no human participants (Section 3.16). The ethics clearance certificate issued by the Unisa School of Computing / College of Agriculture and Environmental Sciences research ethics review committee is reproduced below.

Ethics clearance certificate

Reference: [Unisa SoC/CAES ethics clearance ref: TBD]

Date of issue: [TBD]

(Scanned certificate to be inserted here.)